

IR Sensor and Relay Actuators

Definition · Working Principle · Applications

Submitted by

Pavishna Dhaniya R

Roll No: 23BTT030

PART I: Infrared (IR) Sensor

1. Definition

An Infrared (IR) Sensor is an electronic device that detects and measures infrared radiation — electromagnetic radiation with wavelengths ranging from 700 nm to 1 mm, lying just beyond visible red light in the spectrum. IR sensors operate on the principle that all objects above absolute zero emit or reflect infrared radiation proportional to their temperature and surface properties.

IR sensor modules used in electronics typically integrate an IR LED (transmitter) and a photodiode or phototransistor (receiver) on a single PCB, along with a comparator circuit to produce a clean digital output signal.

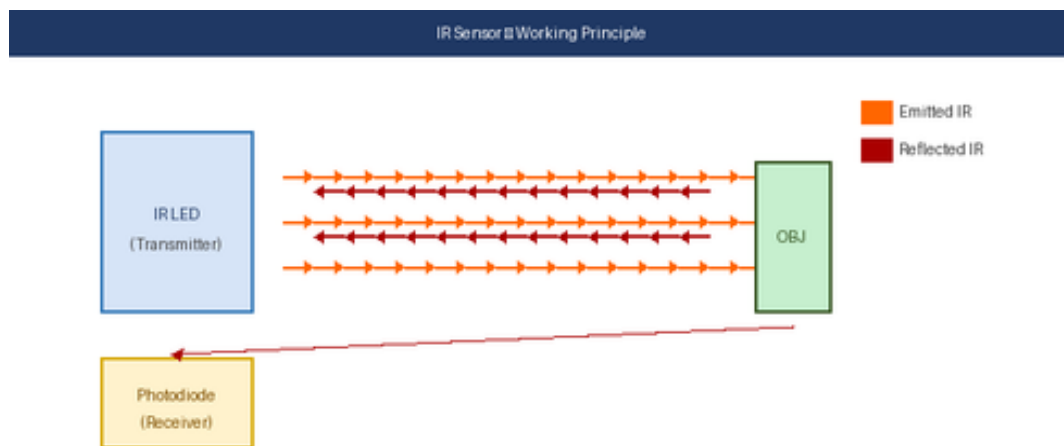


Figure 1: IR Sensor Working Principle — Transmitter and Receiver

2. Working Principle

The IR sensor works on a simple transmitter-receiver principle. The IR LED continuously emits infrared light at approximately 38–940 nm wavelength. When this emitted beam strikes a nearby object, a portion of the radiation is reflected back towards the photodiode receiver.

The intensity of the reflected signal depends on the distance to the object and its surface reflectivity. The photodiode converts the incoming IR photons into an electrical current, which is then fed into a comparator (typically LM393). The comparator compares this signal against a reference threshold set by a potentiometer, outputting a LOW signal (0V) when an object is detected and HIGH (5V / 3.3V) otherwise.

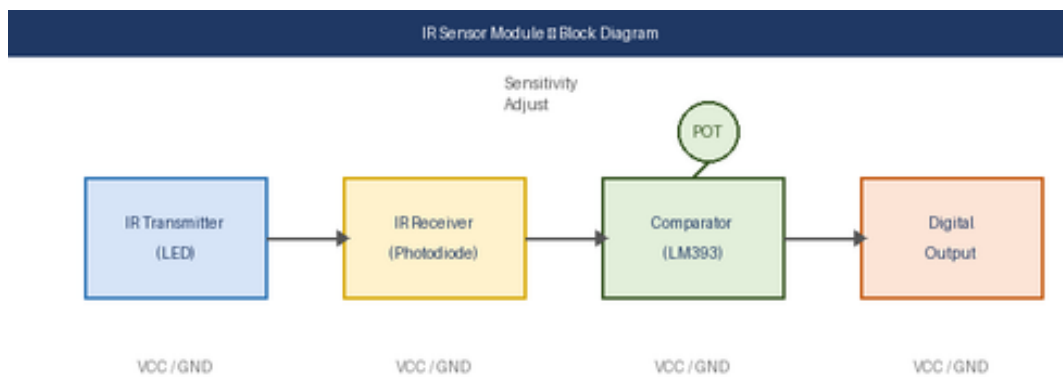


Figure 2: IR Sensor Module Block Diagram — Signal Processing Chain

3. Key Technical Parameters

Parameter	Typical Value / Description
Operating Voltage	3.3V – 5V DC
Detection Range	2 cm – 30 cm (adjustable via potentiometer)
Output Type	Digital (TTL-compatible: HIGH / LOW)
IR Wavelength	850 nm – 940 nm
Response Time	< 2 ms
Operating Temperature	-25°C to +85°C

4. Applications of IR Sensor

- Obstacle and object detection in mobile robots and automated guided vehicles (AGVs)
- Line-following robots — detecting the contrast between a white surface and black line
- TV and consumer electronics remote control signal reception
- Proximity sensing in smartphones to turn off the display during calls
- Automatic doors and faucets in public places using proximity detection
- Burglar alarm systems and perimeter security

- Industrial conveyor belt counting and sorting systems
- Flame detection sensors for fire suppression systems

PART II: Relay Actuator

1. Definition

A relay is an electrically operated switch — an electromechanical actuator that uses a low-power control signal (typically 3.3V–12V DC from a microcontroller) to switch a separate, often high-power circuit carrying AC or DC loads of up to several hundred volts and amperes. The relay provides complete electrical isolation between the control circuit and the load circuit.

Relays are classified by their coil voltage, contact configuration (SPDT, DPDT), and load rating. Common types include electromagnetic relays, solid-state relays (SSR), reed relays, and latching relays.

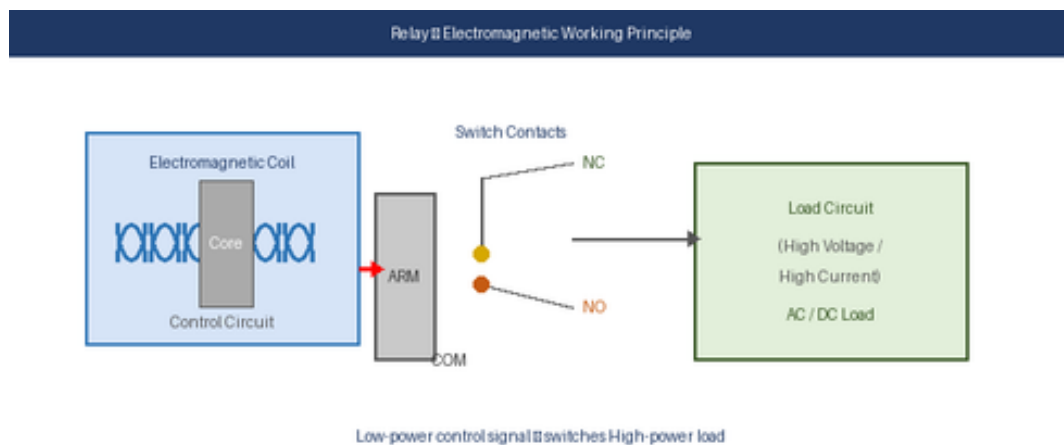


Figure 3: Electromagnetic Relay — Construction and Working Principle

2. Working Principle

When current flows through the relay coil, it generates a magnetic field that magnetises the iron core. This magnetic force attracts the spring-loaded armature (movable arm) towards the core, causing the mechanical contacts to switch position. Specifically:

- Normally Open (NO) contact closes when the coil is energised — current flows through the load.
- Normally Closed (NC) contact opens when the coil is energised — current path to load is broken.
- Common (COM) is the terminal shared between NO and NC contacts.

When the coil current is removed, the magnetic field collapses and a return spring restores the armature to its original position, returning contacts to their default state. A flyback diode is placed across the coil to suppress the voltage spike generated by the collapsing field, protecting the driving transistor or microcontroller GPIO.

3. Key Technical Parameters

Parameter	Typical Value / Description
Coil Voltage	5V / 12V DC (drives the electromagnet)
Contact Voltage Rating	Up to 250V AC / 30V DC
Contact Current Rating	10A – 30A depending on relay type
Contact Configuration	SPDT (Single Pole Double Throw) typical
Pick-up / Drop-off Voltage	~75% / ~10% of rated coil voltage
Operating Temperature	-40°C to +70°C

4. Applications of Relay Actuator

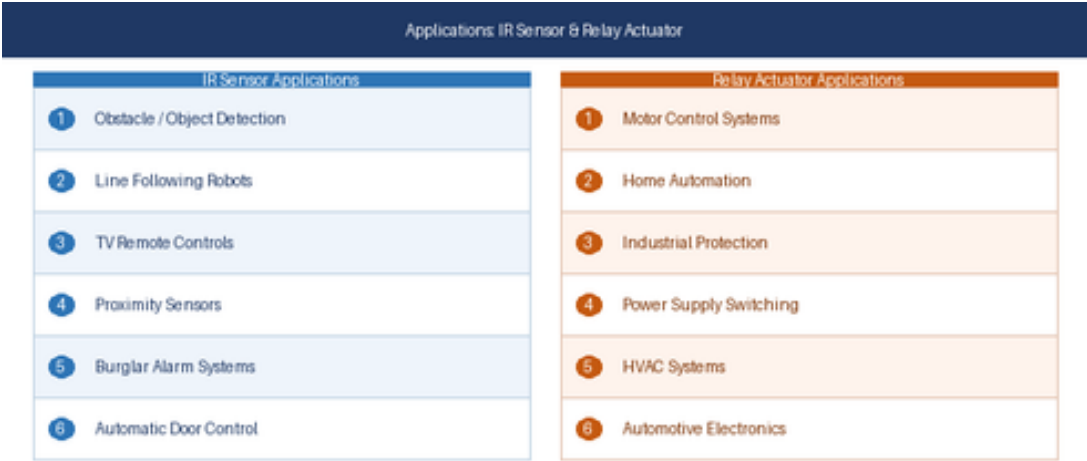


Figure 4: Comparative Application Overview — IR Sensor vs Relay Actuator

- Home automation — remotely switching lights, fans, and appliances via microcontrollers
- Industrial motor control — starting/stopping AC motors using PLC relay outputs

- Overcurrent protection in power distribution panels and circuit breakers
- HVAC (Heating, Ventilation, Air Conditioning) system control and compressor switching
- Automotive electronics — horn, headlight, starter motor, and wiper control relays
- Medical equipment — isolating sensitive electronics from high-voltage supplies
- Solar power systems — battery management and load switching
- Elevator control systems — floor selection and door motor operation

Combined Use: IR Sensor + Relay in a System

In practice, IR sensors and relay actuators are frequently combined to build automated control systems. A classic example is an automatic security light: the IR sensor detects a person approaching, and its digital output triggers a relay module (via a transistor or optocoupler) which in turn switches on a 230V AC lamp. This architecture separates the low-voltage sensing circuitry from the high-voltage load, ensuring both safety and isolation — a fundamental principle in embedded systems and industrial automation.

Conclusion

IR sensors offer a low-cost, reliable, and non-contact method of detecting objects, proximity, and line contrast, making them indispensable in robotics, consumer electronics, and automation. Relay actuators bridge the gap between microcontroller logic levels and real-world high-power loads, providing essential electrical isolation and the ability to control AC and DC circuits. Together, these two components form the backbone of numerous modern embedded and industrial control applications.